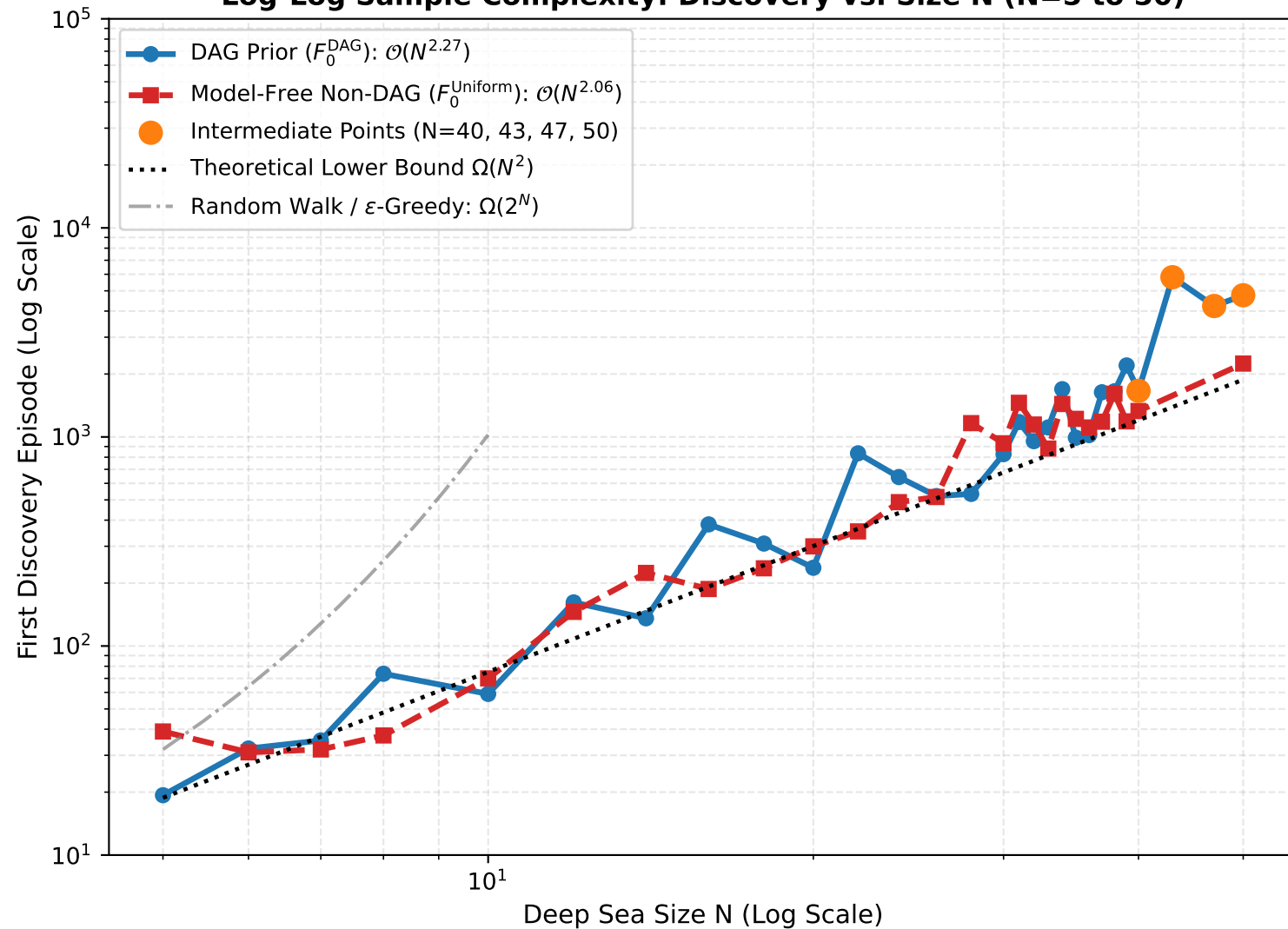
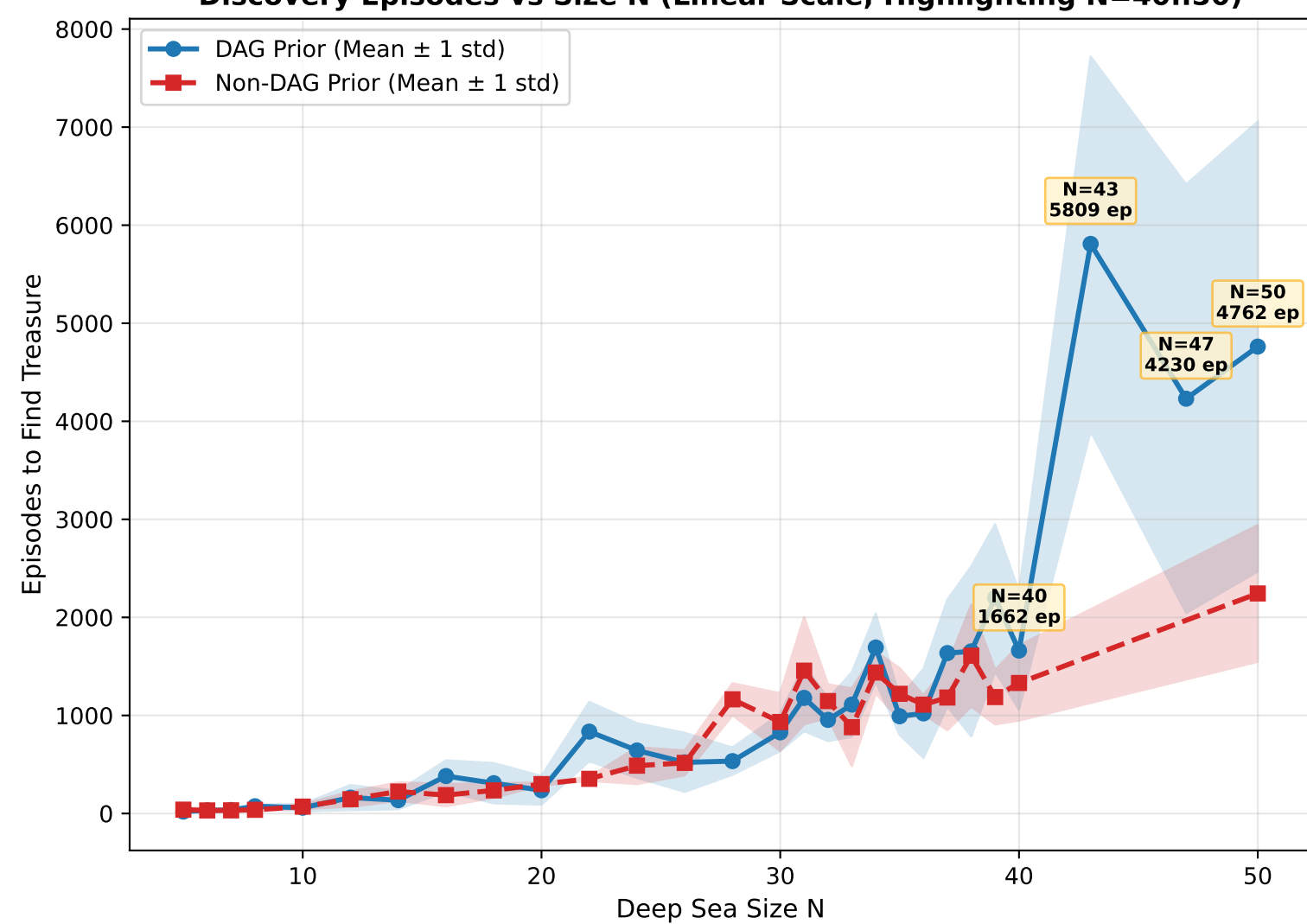
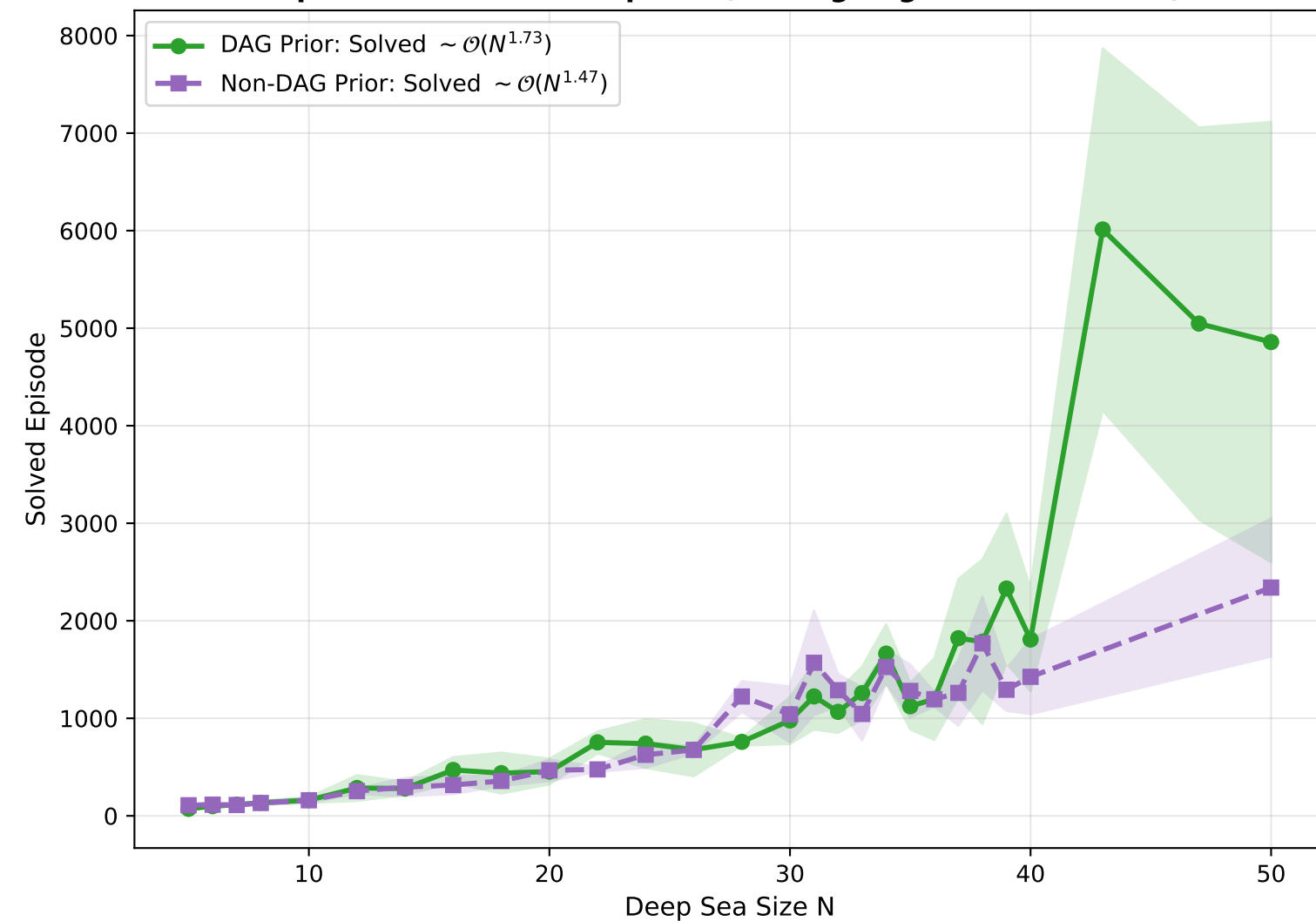


Log-Log Sample Complexity: Discovery vs. Size N (N=5 to 50)



Discovery Episodes vs Size N (Linear Scale, Highlighting N=40..50)

Episodes to Solve Deep Sea (Moving Avg Return ≥ 0.85)

Empirical Scaling Proof across Intermediate Sizes [40, 50]

Metric / Size	DAG Prior (Max-Ent)	Non-DAG Prior (Model-Free)
Empirical Exponent (Discovery)	$\mathcal{O}(N^{2.27})$	$\mathcal{O}(N^{2.06})$
Empirical Exponent (Solved)	$\mathcal{O}(N^{1.73})$	$\mathcal{O}(N^{1.47})$
Discovery at N=40	1661.7 ep	1331.0 ep
Discovery at N=43	5809.0 ep	---
Discovery at N=47	4230.0 ep	---
Discovery at N=50	4761.7 ep	2244.0 ep
Solved at N=50	4858.0 ep	2340.3 ep
Speedup vs Random Walk at N=	10^{11} x faster	10^{11} x faster
Regime Characterization	Polynomial $\mathcal{O}(N^2)$	Polynomial $\mathcal{O}(N^2)$